LUKE BUONANNO

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PROFESSIONAL PROFILE

Aspiring Mechanical Engineer combining hands-on CAD and product design experience with programming, embedded systems, and hardware troubleshooting skills. Builds independent hardware and firmware projects while applying strong analytical thinking and systematic problem-solving across multidisciplinary engineering challenges. Brings a collaborative mindset, effective communication, adaptability, and leadership experience, with a proven ability to work well on teams, resolve challenges constructively, and quickly learn new technologies.

EDUCATION

King's College, Wilkes-Barre, PA *Anticipated Graduation: May 2028*

Bachelor of Science in **Mechanical Engineering** with minors in **Computer Science** and **Mathematics**

Honor's Program | Dean's List: Fall 2024, Spring 2025

TECHNICAL SKILLS

CAD & Design: Autodesk Inventor (advanced, 3D design & drafting), AutoCAD, SolidWorks, Cura slicer,

Programming: Python, C++, Java, HTML5, Discord.js / Discord.py / Node.js; strong comprehensive project development, debugging, and logic/systems planning across languages

Hardware & Lab: LabVIEW, Arduino IDE, multimeter/oscilloscope use, schematic reading, breadboarding, soldering

Tools & Platforms: Excel, MATLAB, Git, Linux, Windows, Microsoft Office, VS Code, Eclipse

Languages: Intermediate Spanish, Beginner Italian

ENTREPENEURSHIP

Lemony Studios, Founder & Engineer, **Macungie, PA** *May 2024 – Present*

- Designed embedded hardware for customers around ESP32/RPi with displays, sensors, power management, etc.
- Engineered CAD assemblies and 3D-printed prototypes from concept through testing and design iteration.
- Developed embedded firmware, user interfaces, networking, and hardware validation software for custom devices.
- Built and maintained a portfolio website showcasing embedded systems, CAD, firmware, and engineering projects.
- Advised a University of Pittsburgh researcher on simplifying a diamond NV quantum sensor sample-access mechanism.

PROJECTS

HenHacks Hackathon, University of Delaware, In-Person *March 2025 and 2026*

- Collaborated on teams to design, build, and present two projects, “Dose Doodle” (a prescription medicine assistant) and “Hen-tersection,” (the traffic intersection reimagined for pedestrians) each within a 24-hour hackathon timeframe (https://devpost.com/LukeBuonanno)

PROFESSIONAL EXPERIENCE

King's College Resident Life, *Resident Assistant,* **Wilkes-Barre, PA** *August 2022 – Present*

- Supported and mentored 13–25 first-year residents, mediating conflicts and enforcing Residence Life policies.
- Planned and led community events that strengthened resident engagement, inclusion, and peer connections.

Wegmans Food Markets, *Front End Employee***, Allentown, PA** *August 2022 – Present*

- Processed customer transactions with 20% faster-than-average checkout speed while maintaining transaction accuracy.
- Delivered exceptional customer service, earning 75+ management commendations for customer satisfaction

IITS, King's College, *IT Student Support Specialist* **, Wilkes-Barre, PA** *October 2024 – March 2025*

- Resolved hardware, software, networking, and printer issues for students and faculty with rapid response times.

St. Luke's Hospital, *Volunteer Patient Advocate,* **Palmerton, PA** *May 2022 – May 2024*

- Supported patients by offering companionship and providing comfort items such as drinks and blankets upon request

ACTIVITIES, LEADERSHIP & AWARDS

- Vice President, Engineering Club, King's College
- Vice President, Coding Club, King's College
- Magnificent Monarch Award, King’s College, May 2026